The Impact of Gamification on College Algebra Performance in Higher Education

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Abstract

College Algebra is widely perceived as difficult and demotivating in higher education, prompting interest in gamification to improve engagement and outcomes. Recent university-level evidence reports performance gains but mixed effects on autonomous motivation and self-efficacy, underscoring the importance of careful game-element design. We conducted a quasi-experimental, mixed-methods study across two intact sections of College Algebra (gamified: $n = 60$; control: $n = 58$), taught in the same semester by the same instructor. The intervention embedded narrative challenges, timed missions, immediate feedback, and progress indicators alongside regular lectures. Outcomes included course assessments (attendance, assignments, midterm, final, total), and post-course perceptions; inferential statistics and thematic analyses were used. Both groups improved from midterm to final (paired t-tests, $p < 0.001$ in each group), with no statistically significant between-group differences in midterm or final grade. Detailed comparisons across all graded components likewise showed no significant advantage for either condition. Nevertheless, engagement indicators favored the gamified section (higher attendance and participation), and most students in the gamified class reported greater motivation, enjoyment, confidence, reduced math anxiety, and better engagement.

In this setting, gamification did not translate into superior summative performance relative to traditional instruction; however, it was associated with improved participation and positive affective responses. These findings align with what recent studies in STEM education have reported, when games are used thoughtfully in teaching, they can help students perform better and stay more engaged. However, motivation does not always increase unless the game features are chosen and applied carefully.

Keywords. Gamification; College Algebra; Higher Education, Engagement.

1. Introduction

Higher education students often perceive mathematics courses as difficult, viewing them as significant barriers to academic success, a perception that frequently originates in secondary school and is reinforced by experiences of anxiety and underperformance Mutegi et al. (2021)**.** College Algebra, a compulsory course for both STEM and non-STEM students who join higher education, plays a crucial role in building analytical and problem-solving skills. However, it is widely regarded as abstract, overly complex, and disconnected from students' fields of study. This perception contributes to low engagement, poor performance and negative attitudes toward mathematics. Traditional lecture-based teaching, while effective for content delivery, often fails to address diverse learning needs or motivate students to participate actively in the learning process.

Gamification has emerged as an innovative strategy to address these challenges. By integrating game elements, such as points, challenges, rewards, and interactive scenarios into instruction, gamification enhances participation, provides immediate feedback, and fosters sustained motivation Liu et al. (2025). This teaching approach is grounded in two well-established theories of learning**;** constructivism and flow theory. By combining these perspectives, the gamified activities transform abstract mathematical problem-solving into a more engaging, goal-oriented, and personally rewarding experience, where students feel both challenged and supported as they progress.

Despite this growing body of evidence, research on the use of gamification in College Algebra, especially in contexts where non-STEM students form a significant proportion of the cohort, remains limited. Recent evidence shows that well-designed gamification can strengthen university students' self-efficacy and engagement in quantitative courses when learning activities are aligned with students' motivational profiles and digital literacy levels Aljamaan et al., (2025). However, more empirical investigations are needed to determine which game design elements are most effective and how these strategies can be tailored to diverse student populations.

This study explores the impact of a gamified intervention in a College Algebra course that serves both STEM and non-STEM majors. By looking at how students' performance, motivation, and attitudes change over time, this study aims to show how well-planned game-like learning activities can make learning basic mathematics more engaging, enjoyable, and effective.

2. Literature Review

2.1Theoretical Background

The integration of games into education is strongly supported by constructivist learning theory and flow theory. According to constructivist theorists such as Piaget (1977) and Vygotsky (1978)**,** students learn best when they actively build knowledge through exploration, collaboration, and reflection, rather than by memorizing information. Learning occurs most effectively when students engage in meaningful tasks that connect new ideas to prior understanding. Gamified environments support this principle by offering interactive tasks, immediate feedback, and opportunities for repeated practice in meaningful contexts.

When students solve problems within an engaging framework, they form stronger conceptual connections and retain content more effectively.

Flow theory, introduced by Csikszentmihalyi (1990), further explains why gamification can be a powerful pedagogical tool. Flow refers to a psychological state of deep immersion and intrinsic motivation that occurs when learners are fully engaged in a task. Flow theory emphasizes that optimal learning happens when tasks are neither too easy nor too difficult, allowing learners to become fully absorbed and intrinsically motivated. Gamified approaches promote this state by providing clear goals, balanced levels of challenge, and instant feedback. In mathematics, a subject often associated with anxiety and perceived difficulty, achieving flow can shift students' experiences from frustration to motivation and sustained engagement Rau et al., (2023). Together, constructivism and flow theory provide a strong theoretical foundation for incorporating game elements into instructional design.

2.2 Gamification in Mathematics Education

Mathematics is frequently perceived as difficult, abstract, and disconnected from students' personal and professional lives. These perceptions contribute to low engagement, high anxiety, and poor performance. Gamification has emerged as a promising approach to address these challenges by integrating game mechanics such as points, leaderboards, challenges, and narratives into mathematics instruction. Research shows that game-based approaches enhance both cognitive and affective outcomes.

A recent meta-analysis revealed that gamification interventions significantly enhanced students' intrinsic motivation and sense of accomplishment by supporting autonomy and relatedness in learning environments Li, Hew, & Du. (2024). Similarly, Gui et al. (2023) found that digital educational games in STEM contexts not only increased engagement but also supported learning by offering adaptive feedback and interactive tasks that made complex concepts more accessible. These studies highlight how gamification can transform abstract problem-solving tasks into interactive, rewarding experiences that build confidence and competence.

Gamification has also been shown to reduce math anxiety, which is particularly important for foundational courses such as College Algebra. By reframing mistakes as part of the learning process and embedding tasks within narratives or competitive frameworks, gamification fosters a growth mindset that encourages persistence and resilience Smiderle et al. (2020). This shift in mindset is critical in helping students overcome negative attitudes toward mathematics and view problem-solving as an achievable and even enjoyable process.

2.3 Gamification in Higher Education

Although gamification gained initial traction in primary and secondary education, its adoption in higher education has accelerated in recent years. It is now widely applied to enhance engagement in challenging courses, including those outside STEM disciplines. A recent quasi-experimental intervention found that implementing leaderboards as part of formative assessments in a higher-education engineering course significantly improved student achievement, although concerns remained about sustaining engagement and intrinsic motivation over time Cigdem, Öztürk & Karabacak. (2024)

Kirkpatrick and Goldberg (2022) found that incorporating game elements in corequisite College Algebra classes improved self-efficacy and engagement, particularly among first-generation and female students. Their findings underscore the importance of tailoring gamified strategies to diverse student populations, including non-STEM majors who often take College Algebra as a general requirement.

Personalization has also been shown to influence the effectiveness of gamification. Oliveira et al. (2022) found that personalized game elements aligned with individual gamer profiles improved flow, enjoyment, and motivation in educational settings. Similarly, Brambilla et al. (2024) highlighted that generic gamification strategies may not uniformly enhance engagement, emphasizing the need for targeted design that accounts for learner diversity.

2.4 Research Gaps

Despite these promising findings, significant gaps remain in the application of gamification to foundational mathematics courses such as College Algebra, particularly in contexts where both STEM and non-STEM students are enrolled. Much of the existing research focuses on other STEM disciplines or younger learners, leaving limited empirical evidence on how gamification can be optimized for diverse higher education cohorts. Moreover, studies such as those by Rodrigues et al. (2022) highlight the novelty effect, where initial enthusiasm wanes if the design lacks meaningful, sustained elements. This underscores the importance of aligning gamified tasks with clear learning objectives, providing adaptive challenges, and integrating feedback that supports both performance and motivation over time.

The present study addresses these gaps by examining the effects of a narrative-driven, blended gamification model in a College Algebra course that serves students across multiple disciplines.

By analyzing performance outcomes and affective measures such as motivation, confidence, and anxiety, this research offers practical insights for educators seeking to implement scalable, evidence-based gamification strategies in higher education.

3. Methodology

3.1 Research Design

This study adopted a quasi-experimental, mixed-methods design to evaluate the influence of gamification on student performance and engagement in a College Algebra course. The quasi-experimental approach was chosen because random assignment of students into treatment and control groups was not feasible within the university's scheduling system. At the institution where this study was conducted, students select course sections based on their personal schedules, availability, and prerequisite completion. Consequently, intact class sections were used, with one section designated as the gamified group and the other as the control group.

Although the absence of randomization limited the experimental purity of the study, methodological rigor was maintained by ensuring that both sections were taught by the same instructor, covered identical content, used the same assessments, and were delivered during the same semester and time slot. To enhance internal validity, pre- and post-test measures were incorporated, along with triangulation of quantitative and qualitative data.

The mixed-methods design allowed for a comprehensive analysis of the intervention by integrating quantitative performance metrics with qualitative data from student surveys and open-ended feedback. This approach ensured a nuanced understanding of both measurable learning outcomes and students' perceptions of the gamified experience.

3.2 Gamification Intervention

The intervention was implemented through a web-based platform that was integrated into the regular College Algebra curriculum. The platform provided interactive lessons, guided practice, and assessments, enhanced with game elements such as points, progress indicators, time-based challenges, and narrative-driven activities to foster engagement and motivation Alt. (2023). Throughout the semester, students participated in themed activities designed to align directly with course topics while promoting active learning and persistence.

3.2.1The gamification design

The gamification design was guided by principles from Flow Theory and Constructivist learning principles**.** Activities were created to keep students both interested and actively involved in their own learning. Features such as having clear goals like finishing a mission or defeating a dragon along with instant feedback through points, progress bars, and accuracy indicators, helped students stay focused and motivated. These elements worked together to create a learning experience where students could lose themselves in the task, enjoy the process, and learn through active participation and problem-solving.

One of the core activities, known as the Set Explorer Mission, required students to work collaboratively to solve set theory problems. Progress was tracked through a points and levels system, and students advanced through a narrative-driven storyline as they successfully completed each challenge. This activity specifically leveraged constructivist principles by promoting collaborative learning and active problem-solving in a structured narrative, while the clear progression and immediate feedback fostered a Flow state. In the Fraction Frenzy Race, students competed against the clock to simplify and multiply rational expressions. This activity encouraged fluency in algebraic manipulation while helping students develop time management skills under pressure, aligning with Flow Theory through its timed challenges and immediate performance feedback, which helped balance skill and challenge. The Equation Escape Room activity required students to solve linear and quadratic equations to collect keys and "escape the room," reinforcing algebraic problem-solving in a highly engaging context that simulated problem-solving under constraints. This design fostered constructivist learning by requiring active application of concepts in a problem-based scenario and supported Flow through clear objectives and progression. Another popular feature, Quadratic Quest: Dragon Slayer, tasked students with applying their knowledge of quadratic equations to defeat a virtual dragon. Accurate solutions reduced the dragon's health, while incorrect attempts partially restored it, encouraging accuracy, persistence, and strategic problem-solving. This activity explicitly applied Flow Theory by providing immediate, tangible feedback (dragon's health), balancing skill and challenge, and offering clear goals, simultaneously promoting constructivist learning through applied knowledge.

Assignments and practice exams were also delivered through the platform and incorporated similar game mechanics, such as progress bars and hints, to keep students motivated as they prepared for assessments. These low-stakes opportunities for consolidation ensured that students had ample practice before high-stakes evaluations. Importantly, the gamification strategy was not designed to replace traditional instruction but to complement it. Weekly lectures and in-class discussions introduced and reinforced concepts, while the platform provided interactive practice that made abstract content more accessible and meaningful. This blended approach ensured that the intervention supported different learning styles and allowed students to engage with the material in multiple ways.

Toward the end of the semester, a practice exam was released through the platform, enabling students to consolidate learning across all units in a low-stakes environment before the final evaluation. Both assignments and the practice exam also included game-like features such as progress indicators, Check Answer buttons that provided immediate validation and Show Hint buttons that nudged students toward the solution path. These mechanics kept engagement dynamic and interactive, making the experience distinct from traditional static textbook quizzes.

Table 1 – Gamification Intervention Summary

Game / Activity	Target Concept	Mechanics (Key Parameters)	Learning Integration & Theoretical Alignment
Set Explorer Mission	Sets (unions, intersections, complements, Venn diagrams)	Team tasks (4–5), weekly 30-minute labs, 12 missions total. Up to 10 points per task plus 5 bonus points; no penalties. Progress and time-tracking interface.	Aligned with Learn pages; promotes collaborative practice and creative problem-solving under time constraints. Encourages constructivist learning through social interaction and active engagement, while supporting Flow through clear goals, immediate feedback, and visible progress.
Fraction Frenzy Race	Rational expressions (simplify, multiply)	Short timed rounds: 5 expressions, 30 seconds per expression; progress displayed (e.g., “Progress: 0/5”).	Reinforces procedural fluency tied to Lesson and Learn content. Develops time-management skills and sustained attention. Supports Flow by balancing challenge and skill while offering instant feedback to keep students immersed.
Equation Escape Room	Linear and simple quadratic equations	Individual and group formats; 15–20 minute sessions, 1–2 per week; 10 individual and 8 group rooms across the semester; instant feedback with step suggestions.	Mirrors sequential problem-solving from Learn pages while alternating autonomy and collaboration. Strengthens constructivist learning through real-world problem contexts and reflection. Promotes Flow by providing clear objectives, visible progression, and a sense of control.
Quadratic Quest: Dragon Slayer	Quadratic equations (factoring, completing the square, quadratic formula)	Narrative “boss-battle” design: Dragon Health = 5.0/5; each correct answer −1 HP, incorrect +0.5 HP; victory when HP = 0.	Provides applied practice and persistence through reward-penalty dynamics. Aligns with Flow Theory via immediate feedback and balanced challenge, and with Constructivism by encouraging knowledge application in a simulated environment.

Graphing Assignment & Practice Exam	Graphing and cumulative review	Digital submission with Check Answer and Show Hint options; includes progress indicators.	Encourages self-evaluation and concept reinforcement. Facilitates Flow through clear progress cues and instant feedback, and supports constructivist learning through guided exploration and opportunities for self-correction.

3.2.2 Control Condition (Traditional Lecture Group)

For comparative analysis, a control section of College Algebra followed a traditional, non-gamified instructional approach**.** The same instructor taught both sections using identical syllabi, learning outcomes, assessment schedules, and instructional materials. Control-group students attended face-to-face lectures emphasizing concept explanation, worked textbook examples, and paper-based exercises without digital timers, points, leaderboards, or interactive feedback. Both groups had access to the same office hours, discussion forums and final assessments; the only distinction was the inclusion of gamified mechanics in the experimental group's digital environment.

3.3 Study Groups

The study was conducted in two sections of the College Algebra course offered during the same semester. The gamified group (MTH 1109D) consisted of 62 students, while the control group (MTH 1109G) included 60 students. The assignment of MTH 1109D as the gamified group was informed by a pre-course survey that assessed students' baseline attitudes toward mathematics, previous gaming experiences, and perceived challenges with the subject. The results revealed that 65% of the students in the gamified group reported negative attitudes toward mathematics, scoring very low on indicators of motivation and enjoyment. Additionally, 75% of the group indicated little or no prior experience with educational games, and 65% were enrolled in non-STEM majors such as International Relations, Psychology, and Journalism. These characteristics made the group particularly relevant for testing the potential of gamification to shift attitudes and engagement levels in a compulsory mathematics course.

3.4 Control Mechanism

To ensure a robust comparative analysis, the control group (MTH 1109G) served as the traditional instructional mechanism. The group was taught using the usual classroom methods without any exposure to the online gamified platform or its interactive features such web-based gamified platform or any of its interactive game elements such as points, progress indicators, time-based challenges, narrative-driven activities. Their learning took place through regular lectures, written assignments, and instructor-led discussions, where feedback was provided after tasks were submitted. By keeping this group separate from the gamified one, the setup ensured that any improvements observed in the experimental group could be confidently linked to the use of gamification rather than other factors.

To maintain methodological rigor, both groups were taught by the same instructor to ensure consistency in content delivery and instructional quality. The classes were scheduled at the same time of day, used the same syllabus, assignments, and grading rubrics, and covered identical topics, ensuring comparability between the groups. The gamified group engaged with the interactive platform activities alongside the traditional lectures, while the control group relied solely on traditional instructional methods, including lectures, paper-based assignments, and instructor-led discussions. This design allowed for an accurate comparison of the effects of gamification without introducing biases related to differences in course content or teaching style.

The full intervention spanned 14 weeks, with game-based tasks integrated into the weekly coursework of the gamified group. The platform tracked engagement and performance analytics, and observations from the instructor provided additional qualitative insights into how students interacted with the gamified elements. By embedding the gamification within the existing structure of the course, rather than presenting it as an external tool, the design ensured that the intervention was a seamless part of the students' learning experience.

3.5 Data Collection

Performance data were extracted from questionnaires issued to study groups. Pre- and post-assessment questionnaires (gamified: n=45; non-gamified: n=37) measured motivation (3 Likert items, composite score 1–5), confidence (4 Likert items, composite score 1–5), and anxiety (3 Likert items, composite score 1–5). Open-ended responses captured qualitative feedback on course experiences, benefits, and challenges of the gamified platform. Questionnaires were piloted for clarity and took 10–15 minutes to complete.

3.6 Data Analysis

Data was cleaned to exclude invalid records. Normality was verified using Shapiro-Wilk tests, and homogeneity of variance was assessed with Levene's test. Independent t-tests compared performance metrics between groups, with Cohen's d for effect sizes. Paired t-tests assessed pre/post changes in motivation, confidence, and anxiety. Qualitative responses were analyzed thematically using NVivo (Braun & Clarke, 2006), with high inter-rater reliability (kappa=0.80). Quantitative and qualitative findings were triangulated.

4. Results

4.1 Performance Metrics

Both groups showed comparable midsemester performance, with the gamified group (MTH 1109D) achieving a mean Midterm Score of 15.42 (SD=3.69) and the control group (MTH 1109G) 15.93 (SD=3.02). Final Exam scores were 21.52 (SD=5.49) for the gamified group and 22.37 (SD=4.74) for the control group. Total Scores were 81.19 (SD=12.36) and 82.88 (SD=11.49), respectively. No significant between-group differences were found (t-tests, p>0.05).

Table 2:Comparative Performance Metrics: Gamified vs. Control Groups

Performance Metric	Gamification (Exp, n=62)	Control (n=60)
Midterm Score (out of 20)	M=15.42, SD=3.69	M=15.93, SD=3.02
Final Exam (out of 30)	M=21.52, SD=5.49	M=22.37, SD=4.74
Total Score (out of 100)	M=81.19, SD=12.36	M=82.88, SD=11.49

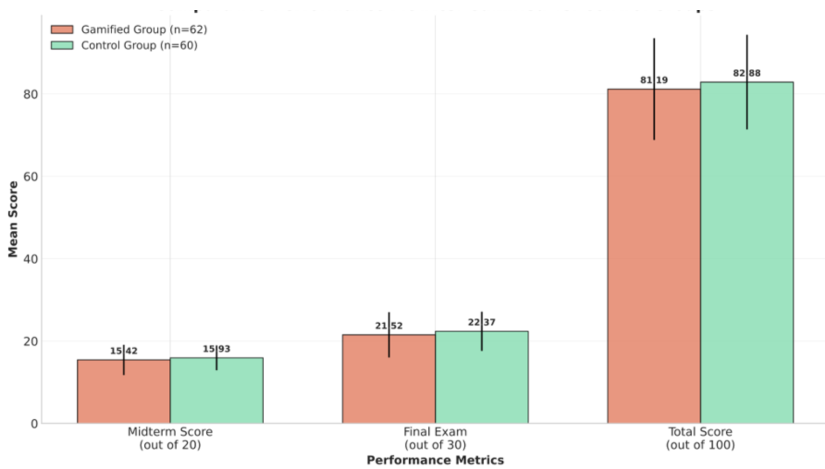

Figure 1:Performance Comparison. Mean scores for midterm (out of 20), final exam (out of 30), and total score (out of 100) for the gamified (n = 62) and control (n = 60) groups, showing no significant differences.

4.2 Affective Outcomes (Gamified Group)

Pre- and post-survey results for the gamified group (n=45) showed significant improvements in motivation, confidence, and anxiety reduction. Motivation increased from 3.2 to 3.8 (d=0.71), confidence from 3.3 to 3.9 (d=0.69), and anxiety decreased from 3.1 to 2.6 (d=-0.64). Paired t-tests confirmed significance (p<0.01). These findings suggest that gamification positively influenced students' emotional experiences, even if academic performance remained comparable to the control group.

Table 1: Pre- and Post-Intervention Affective Outcomes

Measure	Pre Mean (SD)	Post Mean (SD)	Change (d)
Motivation	3.2 (0.9)	3.8 (0.8)	0.71
Confidence	3.3 (0.9)	3.9 (0.8)	0.69
Math Anxiety	3.1 (0.8)	2.6 (0.7)	-0.64

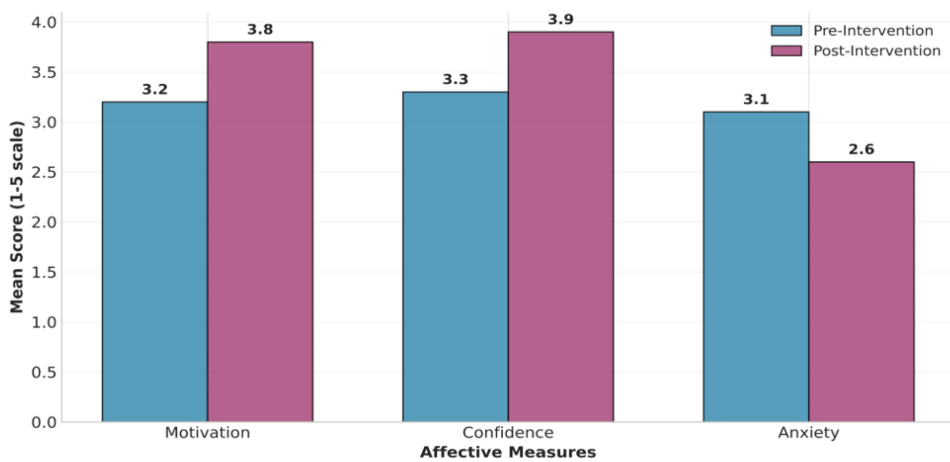

Figure 2: Affective Outcomes. Mean scores (1-5 scale) for motivation, confidence, and anxiety in the gamified group (n = 45), showing improvements in motivation and confidence, and a reduction in anxiety post-intervention.

4.3 Qualitative Feedback

Qualitative data provided additional nuance to these findings. Students frequently described the gamified activities as making mathematics “fun” and “less intimidating,” with several citing the narrative elements and instant feedback as motivators to persist with challenging content. Table 4 summarizes the qualitative feedback themes from the gamified group, highlighting increased motivation and engagement (76%), reduced anxiety (62%), value of instant feedback (67%), and technical challenges (31%). These themes align with Smiderle et al. (2020), who emphasize that the immersive and interactive nature of gamified learning environments can reduce math anxiety and foster a growth mindset.

Table 4: Thematic Analysis of Qualitative Feedback

Theme	Quote	Frequency (%)
Increased motivation and engagement	“It was fun, have never enjoyed maths class this way.”	76%

Reduced anxiety	"It was not as bad as I expected it to be, I am hoping the format will be the same." (Row 8)	62%
Value of feedback	"Instant feedback helped in understanding my mistakes."	67%
Technical challenges	"The platform was sometimes slow, which was frustrating."	31%

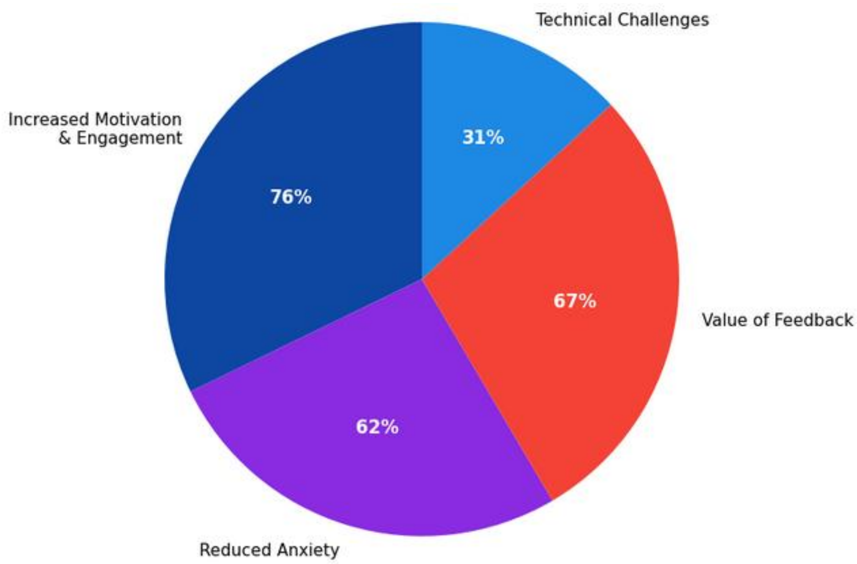

Figure 3: Qualitative Feedback. Distribution of themes from student feedback in the gamified group (n = 45), with increased motivation and engagement (76%), reduced anxiety (62%), value of instant feedback (67%), and technical challenges (31%).

5. Discussion of the results

The findings of this study reveal that the integration of gamification into a College Algebra course positively influenced students' engagement, motivation, confidence, and attitudes toward mathematics, even though it did not result in statistically significant improvements in academic performance. This duality highlights the complex role gamification plays in educational contexts: while it fosters an environment conducive to active learning and reduces negative affective barriers, its direct impact on summative performance may require additional design refinements and alignment with assessment strategies.

The absence of significant differences in total performance scores between the gamified and control groups aligns with findings from Ortiz-Rojas et al. (2025), who reported that leaderboard-based gamification improved engagement but had mixed effects on academic outcomes. These patterns suggest that while gamification creates a more supportive and engaging learning environment, additional scaffolding or integration with formal assessments may be needed to translate these affective benefits into measurable performance improvements.

The significant gains observed within both the gamified and control groups between midsemester and final exam scores indicate that both instructional approaches supported students' academic growth. However, the slightly larger improvement in the control group suggests that traditional methods, which emphasize direct instruction and structured practice, may better prepare students for summative assessments in the absence of deliberate alignment between gamified tasks and test content.

Despite the lack of significant performance differences, the gamified intervention produced clear affective benefits. Students in the gamified group reported higher levels of motivation, enjoyment, and confidence, along with reduced anxiety toward mathematics. These findings echo the conclusions of Alt (2023) and Gui et al. (2023), who found that gamification promotes positive attitudes and sustained engagement by making abstract concepts more accessible and providing immediate, constructive feedback. Such outcomes are particularly valuable in a compulsory course like College Algebra, which serves both STEM and non-STEM students and is often associated with high levels of anxiety and low engagement.

Qualitative data provided additional nuance to these findings. Students frequently described the gamified activities as making mathematics "fun" and "less intimidating," with several citing the narrative elements and instant feedback as motivators to persist with challenging content. These themes align with Smiderle et al. (2020), who emphasize that the immersive and interactive nature of gamified learning environments can reduce math anxiety and foster a growth mindset. The higher participation and attendance rates observed in the gamified group further suggest that students found the approach more engaging, even if this did not immediately translate into higher scores.

The findings also highlight areas for improvement in the design and implementation of gamification. Technical issues and occasional lack of clarity in instructions, reported by some students, suggest that robust platform design and clear instructional guidance are essential for sustaining engagement. Moreover, the absence of platform analytics limited the ability to correlate levels of interaction with performance outcomes. Future research should leverage usage data to better understand the relationship between engagement and achievement, as suggested by Rodrigues et al. (2022), who noted the importance of monitoring long-term patterns to address novelty effects and maintain motivation.

The results underscore that gamification, when thoughtfully integrated into the curriculum, can create a more inclusive and supportive learning environment that benefits diverse student populations. While the immediate impact on academic performance may be modest, the affective gains observed especially for students with negative attitudes toward mathematics or limited prior exposure to educational technology represent an important step toward improving persistence and overall learning experiences in foundational courses.

6. Conclusion and Recommendations

This study investigated the impact of a gamified instructional approach on student engagement, performance, and attitudes in a College Algebra course that serves both STEM and non-STEM students. The findings demonstrate that while gamification did not lead to statistically significant improvements in summative academic performance, it produced substantial affective benefits, including increased motivation, higher confidence, reduced anxiety, and greater participation and attendance. These outcomes highlight the potential of gamification as a pedagogical strategy to create more supportive and engaging learning environments in foundational mathematics education.

The lack of measurable performance gains suggests that the design and implementation of gamified activities require careful alignment with formal assessments to maximize their instructional impact. Oliveira et al. (2022), which emphasize that gamification is most effective when it is deliberately integrated with curriculum objectives and assessment strategies. By ensuring that the skills reinforced through gamified tasks are directly transferable to graded assessments, future interventions may translate affective gains into more pronounced performance improvements.

The study also highlights the importance of addressing technical and design-related challenges. Feedback from students indicated that platform usability and clear instructional guidance are critical for sustaining engagement and ensuring that all learners benefit from the intervention. Moreover, the absence of detailed platform analytics limited the ability to examine correlations between levels of engagement and academic performance. Incorporating analytics in future iterations would provide valuable insights into usage patterns and their relationship to learning outcomes.

Based on the findings of this study, several recommendations are proposed for educators, instructional designers, and future researchers. Gamification should be integrated as a complement to traditional teaching methods rather than a replacement. By leveraging its strengths in fostering motivation, reducing anxiety, and encouraging participation, educators can create more engaging learning environments while maintaining the rigor of structured instruction.

Careful alignment between gamified activities and formal assessments is essential to ensure that improvements in engagement and affective outcomes are translated into measurable academic performance gains.

Platform design should prioritize usability and reliability, including real-time analytics for monitoring student progress and tailoring interventions. Incorporating these features would allow instructors to make data-driven adjustments to support students throughout the semester. Fourth, technical issues that were noted by some students highlight the importance of investing in robust infrastructure and clear user guidance to minimize barriers to participation.

Future research should focus on inclusivity and accessibility. Students with disabilities, such as those with visual impairments, hearing impairments, or mobility challenges, remain underrepresented in gamification research and design. Platforms could integrate accessibility features such as screen reader compatibility, audio-based challenges, tactile or haptic feedback, and voice-command functionality to create equitable learning experiences. Blind or visually impaired students could engage with audio-driven algebra quests or puzzle-solving activities that rely on verbal cues rather than visual displays. Similarly, captioned instructions and adjustable sound settings would better support students with hearing difficulties. Designing accessible gamified tasks not only promotes inclusivity but also ensures that gamification serves as a tool for reducing educational inequities rather than unintentionally reinforcing them.

Further research should explore how inclusive game mechanics impact both performance and engagement for students with diverse learning needs, ensuring that gamification evolves as a universally beneficial pedagogical approach.

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